

Multimodal Retrieval-Augmented Generation for Chart and Document Understanding: A Survey

Multimodal retrieval-augmented generation for chart and document understanding connects large language models to visual, textual, and structured evidence from pages, figures, tables, forms, and scanned documents. This survey reviews the field as an end-to-end pipeline that includes visual document parsing, OCR, layout modeling, chart and table reasoning, cross-modal retrieval, evidence fusion, grounded answer generation, and evaluation. We synthesize an arXiv-derived corpus of 300 papers collected during this run and organize the literature around multimodal RAG architectures, chart question answering, document VQA, table and form understanding, retrievers and indexes, large multimodal models, benchmarks, and faithfulness. The survey argues that chart and document RAG differs from text-only RAG because evidence is distributed across pixels, layout, visual marks, text spans, and latent data values. We close with prospects for provenance-aware visual evidence, layout-grounded citation, robust OCR, chart data recovery, privacy-aware document search, and evaluations that separate retrieval failure from visual reasoning failure.

Keywords:

multimodal retrieval-augmented generation; document understanding; chart question answering; visual document AI; OCR; large multimodal models

1 Introduction

Retrieval-augmented generation (RAG) becomes substantially more complex when the evidence is multimodal. In chart and document understanding, relevant information may appear as words recognized by OCR, layout relationships among blocks, axes and legends in charts, cells in tables, visual marks in figures, or metadata attached to pages. A model must decide what to retrieve, how to represent visual evidence, and how to generate an answer whose claims can be traced back to specific regions, text spans, or data values [61, 72, 85, 121, 126, 188, 194, 284]. This makes multimodal RAG an intersection of information retrieval, document AI, vision-language modeling, chart reasoning, and grounded generation.

Text-only RAG usually treats the retrieved unit as a passage. Chart and document RAG rarely has such a clean unit. A page may contain a title, figure, paragraph, footnote, table, and form field; a

chart may encode values by position, color, shape, and axis scale; and a scanned document may contain OCR noise and reading-order ambiguity. Retrieval can occur over images, page crops, layout nodes, detected text, table structures, chart data, or summaries generated by a multimodal model [22, 61, 85, 121, 126, 188, 194, 284]. The generator then has to reconcile evidence across modalities rather than merely quote passages.

This survey focuses on multimodal RAG for chart and document understanding. We distinguish it from general multimodal question answering by emphasizing external evidence stores, retrieval policies, provenance, and answer grounding. The central question is how to build systems that can find the right visual-document evidence and use it faithfully.

2 Background

2.1 From Document AI to Multimodal RAG

Document AI has long studied OCR, layout analysis, key information extraction, table recognition, form understanding, and document question answering. Multimodal RAG adds a retrieval and generation layer on top of these components. Rather than processing one page in isolation, a system may search across a corpus of PDFs, slides, forms, reports, figures, or spreadsheets, retrieve relevant evidence, and answer in natural language. This setting requires both local perception and corpus-level indexing [61, 72, 85, 126, 188, 194, 232, 284].

2.2 Charts as Visualized Data

Charts are compact visual encodings of data. Understanding them requires recognizing chart type, axes, legends, data marks, labels, scales, and relationships such as trends or outliers. Chart RAG can retrieve a chart, parse its visual structure, recover approximate data values, and ground answers in visual elements. The difficulty is that a chart answer may depend on numerical interpolation, visual comparison, or external context from the surrounding document [50, 72, 85, 188, 194, 232, 278, 300].

2.3 Grounding Across Pixels, Text, and Structure

Grounding in multimodal documents is not only textual attribution. A generated claim may depend on a bounding box, a table cell, a plotted bar, an OCR token, or a relation among layout elements. Reliable systems therefore need multimodal provenance: each answer should identify what evidence was retrieved, where it appeared, how it was transformed, and why it supports the claim [21, 110, 175, 188, 271, 278, 281, 284].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with multimodal retrieval-augmented generation, visual RAG, document understanding, chart understanding, chart question answering, document question answering, DocVQA, ChartQA, InfographicVQA, TextVQA, OCR, layout-aware document modeling, table understanding, document retrieval, image retrieval, visual question

answering, large multimodal models, and visual grounding. Records were retained when title or abstract evidence connected LLMs, language models, multimodal models, or vision-language models with retrieval, chart or document understanding, OCR/layout processing, table reasoning, grounding, evaluation, or robustness. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize methods along six axes. The first is *evidence type*: text spans, OCR tokens, page images, layout blocks, charts, tables, forms, screenshots, or multimodal summaries. The second is *indexing unit*: whole document, page, crop, region, table row, chart element, object, or generated caption. The third is *retrieval signal*: lexical search, dense text embeddings, image embeddings, layout-aware embeddings, cross-modal alignment, or hybrid ranking. The fourth is *reasoning target*: document QA, chart QA, table reasoning, key information extraction, comparison, summarization, or evidence verification. The fifth is *generation control*: citation, bounding-box attribution, structured answer schemas, abstention, and verifier-guided revision. The sixth is *evaluation*: retrieval recall, answer accuracy, chart-data fidelity, citation support, robustness to OCR noise, latency, and privacy.

3.3 Multimodal RAG Architectures

Multimodal RAG systems differ in where they convert visual evidence into text. Some systems caption images, OCR pages, or serialize layout before indexing; others retrieve visual embeddings and let a multimodal model inspect the evidence directly. Hybrid systems keep both representations: textual indexes provide lexical precision, while visual embeddings capture figures, layout, and diagrams. The architectural choice affects what can be cited, how retrieval errors are diagnosed, and whether the generator can verify details against pixels [22, 61, 85, 121, 126, 188, 194, 284].

3.4 Chart Retrieval and Chart Question Answering

Chart-oriented systems must handle a visual grammar rather than only natural language. Retrieval may find a relevant chart from a report, slide deck, or figure collection; parsing then identifies marks, axes, legends, titles, and labels. Answer generation may require comparing values, estimating slopes, aggregat-

ing categories, or explaining trends. A faithful chart RAG system should preserve the link from a numerical answer to the underlying visual mark or recovered data value [50, 72, 85, 188, 194, 232, 278, 300].

3.5 Document Understanding, Layout, and OCR

Document RAG relies on OCR and layout analysis to transform pages into searchable evidence. Reading order, headers, footers, sidebars, tables, stamps, and handwritten marks can all affect retrieval and reasoning. Layout-aware document models and OCR-free vision-language models offer complementary trade-offs: OCR creates explicit text spans and citations, while image-first models can use visual context that OCR may miss. Practical systems often need both [61, 72, 85, 126, 188, 194, 232, 284].

3.6 Tables, Forms, and Structured Evidence

Tables and forms sit between text and databases. They encode meaning through rows, columns, headers, merged cells, keys, and spatial alignment. Retrieval over tables can target captions, headers, cells, or serialized structures; generation may need aggregation, comparison, or unit conversion. Form understanding adds key-value extraction and field grounding. These structured settings benefit from schemas and validators that constrain the generator’s answer space [61, 72, 85, 179, 188, 194, 232, 284].

3.7 Retrievers, Indexes, and Cross-Modal Alignment

The retrieval layer determines whether the generator ever sees the right evidence. Multimodal document corpora can be indexed with OCR text, captions, layout graphs, page thumbnails, region embeddings, table schemas, chart descriptors, or combinations of these features. Cross-modal alignment is central: a text query such as “sales in 2021” may need to retrieve a bar in an image, a cell in a table, or a figure caption. Reranking and query decomposition help when the initial retrieval unit is too coarse [21, 126, 150, 157, 159, 175, 180, 281].

3.8 Large Multimodal Models as Readers and Reasoners

Large multimodal models can inspect retrieved images, reason over layout, read charts, and produce natural-language answers. They reduce the need

to hand-engineer every parser, but they also create new failure modes: hallucinated OCR, misread axes, inconsistent counting, and unsupported citations. A useful design pattern is to pair an MLLM reader with structured evidence extraction and verification, so that visual reasoning is checked against explicit text spans, tables, or recovered chart data [22, 61, 72, 85, 126, 179, 194, 232].

3.9 Benchmarks and Evaluation

Evaluation should disentangle retrieval from visual reasoning. A system may fail because it retrieved the wrong page, because OCR missed a token, because chart parsing misread an axis, or because the generator ignored correct evidence. Benchmarks for DocVQA, ChartQA, table QA, infographic QA, text-rich VQA, and multimodal RAG each stress different parts of the pipeline. Strong evaluations report answer accuracy together with evidence recall, citation faithfulness, visual grounding, robustness to noise, and cost [61, 72, 85, 121, 126, 188, 194, 284].

3.10 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

Multimodal rag architectures and visual evidence fusion The multimodal rag architectures and visual evidence fusion cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [58, 74, 173, 175, 218, 231, 234, 253, 281]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [123, 135, 144, 164, 165, 201, 207, 219, 291]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [9, 81, 100, 127, 145, 181, 238, 265, 280]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [7, 101, 108, 161, 182].

Chart understanding and chart question answering The chart understanding and chart question answering cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [85, 87, 140, 146, 148, 204, 274, 276, 278]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [29, 56, 76, 80, 99, 120, 152, 230, 247]. Additional retained studies broaden the cluster

with variants of interfaces, domains, training signals, and deployment assumptions [77, 258, 288].

Document understanding, layout, and ocr The document understanding, layout, and ocr cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [50, 72, 119, 159, 195, 232, 275, 292, 296]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [39, 40, 47, 73, 90, 92, 115, 197, 205]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [32, 46, 49, 118, 167, 170, 178, 237, 273]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [48, 113, 129, 132, 142, 158].

Table, form, and structured document reasoning The table, form, and structured document reasoning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [21, 55, 82, 131, 179, 188, 194, 284, 300]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [16, 37, 45, 54, 71, 208, 213, 263, 266]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [10, 59, 96, 104, 241, 251, 260, 282, 285]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [26, 31, 34, 86, 95, 168, 190, 233, 279]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [4, 24, 52, 70, 125, 206, 264, 267, 272]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [66, 107, 212].

Retrievers, indexes, and cross-modal alignment The retrievers, indexes, and cross-modal alignment cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [15, 25, 27, 38, 109, 166, 180, 202, 295]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [53, 60, 97, 133, 153, 169, 177, 199, 226]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [44, 63, 84, 130,

138, 139, 209, 223, 290]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [12, 103, 105, 106, 171, 186, 196, 245, 286]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [14, 184, 227].

Multimodal llms and vision-language models The multimodal llms and vision-language models cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [22, 89, 93, 137, 210, 225, 243, 250, 277]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [30, 36, 41, 67, 155, 156, 217, 240, 269]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [43].

Benchmarks, datasets, and evaluation The benchmarks, datasets, and evaluation cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [1, 61, 110, 121, 126, 150, 259, 271, 297]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [18, 19, 111, 122, 157, 172, 211, 216, 257]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [5, 17, 33, 203, 235, 239, 244, 249, 287]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [51, 62, 69, 78, 114, 160, 183, 187, 236]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [2, 6, 28, 75, 124, 229, 248, 256, 293]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [57, 64, 88, 91, 141, 149, 191, 246, 262]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [20, 35, 116, 174, 214, 215, 242, 255, 270]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [23, 83, 94, 98, 112, 220, 224, 254, 268]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [3, 8, 13, 136, 151, 162, 189, 192, 261]. Additional retained studies broaden the

cluster with variants of interfaces, domains, training signals, and deployment assumptions [11, 65, 117, 134, 198, 200, 228, 289, 298]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [68, 79, 102, 147, 163, 221, 222, 283, 294]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [128, 193, 252].

Faithfulness, attribution, and robust document grounding The faithfulness, attribution, and robust document grounding cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [42, 143, 154, 176, 185, 299].

4 Challenges and Prospects

4.1 Evidence Granularity and Provenance

Multimodal RAG needs provenance finer than document-level citation. Users often need to know which chart, axis label, table cell, OCR span, or bounding box supports an answer. Future systems should expose retrieved regions and transform logs so that visual evidence remains auditable after summarization or generation.

4.2 OCR Noise, Layout Ambiguity, and Chart Data Recovery

OCR errors, reading-order mistakes, and chart-data extraction errors can cascade into wrong answers. Robust systems will combine redundant representations: raw pixels, OCR text, layout graphs, captions, table schemas, and generated summaries. Cross-checking among these views can reveal contradictions before final generation [61, 72, 85, 126, 188, 194, 232, 284].

4.3 Faithfulness of Multimodal Generation

Multimodal models can produce confident answers that are not supported by retrieved evidence. Citation and attribution must therefore be evaluated at the claim level. A promising direction is to require generated answers to point to evidence objects, then verify entailment or numerical consistency against those objects [21, 110, 175, 188, 271, 278, 281, 284].

4.4 Privacy and Governance of Document Corpora

Document collections often contain private, regulated, or business-sensitive data. Multimodal RAG systems need access control, redaction, retention policy, and query logs that protect visual as well as textual information. Privacy is especially hard when sensitive content appears in screenshots, scans, or handwritten annotations.

4.5 Toward Unified Visual-Document Memory

The long-term prospect is a document memory layer that indexes text, layout, visual marks, tables, and provenance in one searchable representation. Such a system would support grounded question answering, comparison across reports, chart data extraction, and auditable summarization without forcing every document into plain text.

5 Conclusion

Multimodal RAG for chart and document understanding extends retrieval-augmented generation from passages to rich visual documents. This survey reviewed architectures, chart QA, document layout and OCR, tables and forms, retrievers, large multimodal readers, benchmarks, and faithfulness. The central lesson is that reliable document RAG requires both good retrieval and good evidence representation. Future systems should preserve provenance across pixels, text, layout, and structure so that answers are not only plausible but grounded in inspectable visual evidence.

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